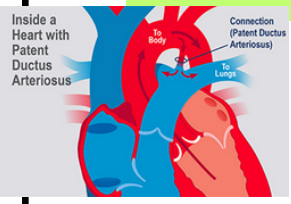
What medication is given to keep the PDA open? What medication is given to close the PDA?

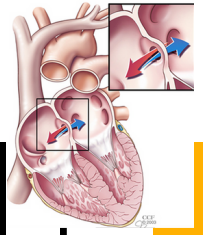
Left Sided HF (s/s?)

Acyanotic

Patent Ductus Arteriosus

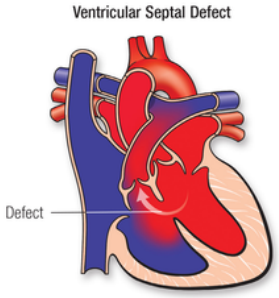


- Aorta–pulmonary artery connection; left-to-right shunt.
- Open: prostaglandin E1 if duct-dependent.
- Close: ibuprofen/indomethacin in selected preterm infants; catheter/surgery.



Atrial Septal Defect

- Opening between atria → usually left-to-right shunt.
- Often asymptomatic; murmur/fixed split S2.
- Larger defects may need closure.
- Left-sided HF clues: tachypnea, crackles, feeding fatigue and pulmonary congestion.

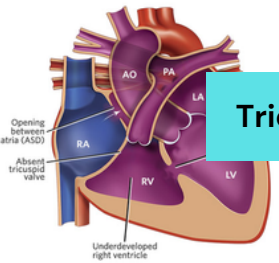


Ventricular Septal Defect

- Opening between ventricles → usually left-to-right shunt.
- Murmur; large shunt causes pulmonary overcirculation, tachypnea, poor feeding/growth.
- Small defects may close; HF treatment and repair when indicated.

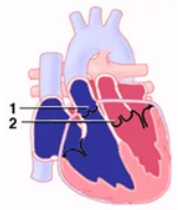
Ductal Dependent

Cyanotic



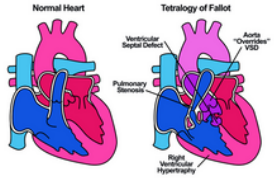
Tricuspid Atresia

- Absent tricuspid opening restricts right-heart flow.
- Survival requires mixing and a route to lungs.
- Cyanosis; prostaglandin if duct-dependent.
- Staged surgical palliation.



Transposition of the Great Arteries (TGA)

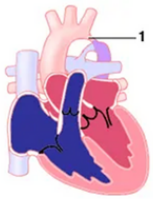
- Aorta from RV; pulmonary artery from LV → parallel circuits.
- Blood must mix through ASD/VSD/PDA.
- Severe neonatal cyanosis.
- Prostaglandin, possible atrial septostomy; arterial switch repair.



Tetralogy of Fallot

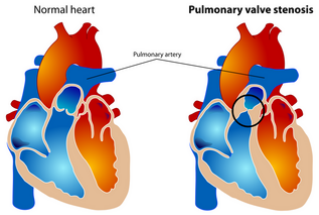
- Four: VSD, pulmonary outflow obstruction, overriding aorta, RV hypertrophy.
- Cyanosis depends on obstruction.
- Hypercyanotic spell: calm child, knee-chest position, urgent help/O2 and ordered treatment.
- Repair; prostaglandin only when pulmonary flow is duct-dependent.

Obstructive Blood Flow



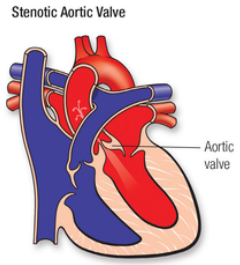
Coarctation of the Aorta

- Narrow aorta → high arm BP, weak femoral pulses, lower leg BP.
- Compare upper/lower extremity pulses and pressures.
- Critical newborn: shock as duct closes.
- Prostaglandin if duct-dependent; catheter/surgical repair.



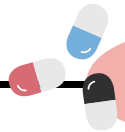
Pulmonary Stenosis

- Obstruction from RV to pulmonary artery.
- RV workload rises; murmur, exertional symptoms.
- Severe disease can cause cyanosis.
- Balloon valvuloplasty or surgery; critical newborn may need prostaglandin.



Aortic Stenosis

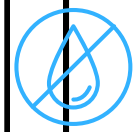
- Obstructed LV outflow; increased LV workload.
- Murmur; exertional syncope/chest pain are concerning.
- Critical newborn may develop low output/shock.
- Activity advice and catheter/surgical treatment by cardiology.



Cardiac Medications

- Diuretics: reduce congestion; check weight, electrolytes and urine.
- ACE inhibitors: reduce afterload; monitor BP, renal function and potassium.
- Digoxin: prescribed contractility support; apical pulse and toxicity monitoring.
- Prostaglandin E1: maintains ductal flow; monitor apnea and BP.
- Use individualized pediatric dose/hold parameters.







Nursing Interventions for HF

- Small, frequent high-energy feeds; rest breaks.
- Daily weight; strict intake/output.
- Monitor work of breathing, perfusion and oxygenation.
- Cluster care; oxygen as prescribed.
- Follow fluid/sodium and medication plan; teach worsening signs.





Cardiac Catheterization

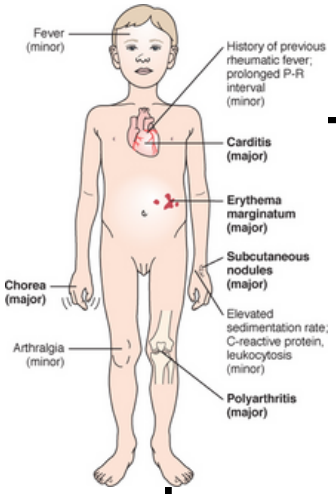
- Before: consent, fasting, baseline pulses and allergies.
- After: monitor access site, bleeding, distal pulses/color/temp.
- Keep affected limb positioned as ordered; check vitals and urine.
- Report bleeding or impaired limb perfusion immediately.



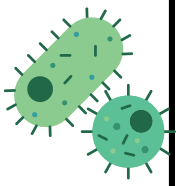
Kawasaki Disease

- Systemic vasculitis; coronary artery aneurysm risk.
- Persistent fever with red eyes, red/cracked lips or strawberry tongue, rash, extremity changes and cervical node enlargement.
- Incomplete presentations also occur.
- Echocardiography and inflammatory labs support evaluation.
- IV immunoglobulin + aspirin under specialist direction.
- Monitor cardiac status, hydration and response.
- Follow cardiology/echo plan; discuss live-vaccine timing after IVIG.

VS



Rheumatic Fever



- Immune complication after group A streptococcal infection.
- Major Jones manifestations: carditis, migratory polyarthritis, chorea, erythema marginatum, subcutaneous nodules.
- Diagnosis uses criteria plus evidence of preceding infection.
- Eradicate strep; anti-inflammatory treatment and cardiac care.
- Long-term antibiotic prophylaxis may prevent recurrence.
- Teach completion of antibiotics and follow-up.
- Main concern: lasting valve damage.